Solar System Models · Staff Answers

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Arrival Answers

1. Mercury.
2. About 365 days (1 year).
3. Jupiter (12 Earth years).
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Planets further from the Sun travel a much longer path and move more slowly in orbit.

Hinge: A. Following the pattern, Saturn's orbit time (29 Earth years) sits between Jupiter (12 years) and Uranus (84 years).

Evidence Desk Cards

Mercury: 58 million km, orbit = 88 days: Inner planet data: shortest distance from the Sun and shortest orbit time.

Earth: 150 million km, orbit = 365 days (1 year): Inner planet data: 1 Earth year is 365 days at 150 million km.

Jupiter: 778 million km, orbit = 12 Earth years: Outer giant data: over 5 times further than Earth, taking 12 years per orbit.

Neptune: 4,500 million km, orbit = 165 Earth years:

Outer giant data: furthest planet, taking 165 Earth years for one orbit.

Shows that inner planets cluster close while outer giants spread far apart: Model strength: makes the huge spacing gap beyond Mars easy to see.

Planet dots on the strip are much too large for the 1 m distance scale: Model limitation: distance and planet diameter are not drawn to the same scale.

Check & Repair Answer

Wall posters space planets evenly just to fit labels; our scaled data shows huge gaps between the outer giants.

Exit Ticket Answers

1. Neptune (165 Earth years). 2. As distance from the Sun increases, the time taken for one complete orbit increases. 3. Strength: shows the true relative spacing (inner 4 bunched, outer 4 wide apart). Limitation: planets are shown stationary in a line, and planet dot sizes are not to the same scale.